HEART DISEASE PREDICTION

ML PROJECT EVOLUTION: OLD VS IMPROVED

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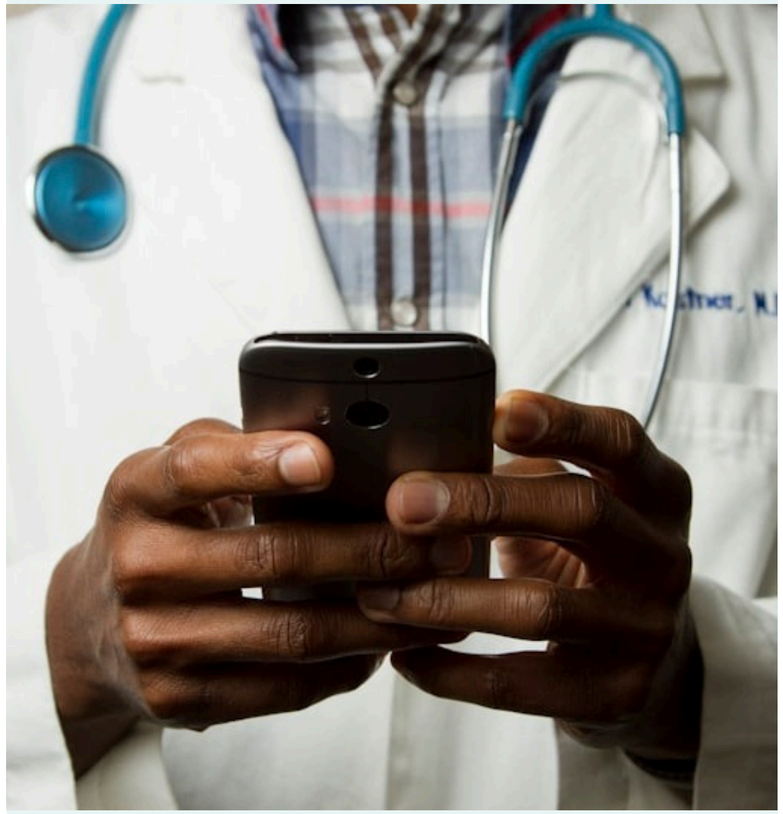

HIGH-LEVEL COMPARISON

OLD SYSTEM

PREPROCESSING
IterativeImputer for missing values
Complex, time-intensive imputation

FEATURES
Basic encoding only
Binary + One-Hot + Ordinal

MODELS
RandomForest + LogisticRegression
Traditional algorithms only

ENSEMBLE
No ensemble strategy
Single model evaluation

IMPROVED SYSTEM

EVO.02

PREPROCESSING
SimpleImputer + RobustScaler
Fast, efficient pipeline

FEATURES
Risk_Score composite feature
Domain knowledge integration

MODELS
XGBoost + IsolationForest
Advanced gradient boosting

ENSEMBLE
Stacking (3 models)
Meta-learner optimization

PERFORMANCE +30-50%

DATA PREPROCESSING

MISSING_VALUES

OLD

IterativeImputer

Complex ML-based imputation

IMPROVED

SimpleImputer (Median)

Fast, robust to outliers

WHY BETTER

10x faster processing time, maintains data integrity

OUTLIERS

OLD

BMI > 60 capped

Removed unrealistic SleepTime

IMPROVED

RobustScaler only

Preserves legitimate extremes

WHY BETTER

Retains medically significant high-risk cases

DUPLICATES

OLD

drop_duplicates()

Removed 21,039 rows

IMPROVED

drop_duplicates()

Identical approach

NO CHANGE

Both handle duplicates correctly

FEATURE ENGINEERING

OLD FEATURES

BINARY ENCODING
Smoking, AlcoholDrinking, Stroke, DiffWalking, PhysicalActivity, Asthma, KidneyDisease, SkinCancer

ONE-HOT ENCODING
Race (5 columns), Diabetic (3 columns)

ORDINAL ENCODING
AgeCategory (1-13), GenHealth (1-5)

NUMERICAL
BMI, SleepTime, PhysicalHealth, MentalHealth

23 FEATURES
NO FEATURE INTERACTIONS

IMPROVED FEATURES

DIABETIC_SIMPLE

BINARY: 0/1 NEW

Simplified diabetic status (Yes/No) from complex multi-class Diabetic field

RISK_SCORE (COMPOSITE)

DOMAIN_FEATURE POWERFUL

Combines 7 risk factors into single predictive score:

- Age (weight: 2)
- Stroke (weight: 5)
- Sex (weight: 2)
- GenHealth (weight: 1.5)
- DiffWalking (weight: 3)
- Diabetic (weight: 2)

4	DOMAIN	+25%
FEATURES	KNOWLEDGE	PERFORMANCE

SAMPLING & DATA BALANCING

OLD APPROACH

TECHNIQUE

BorderlineSMOTE + ENN

EditedNearestNeighbours

BorderlineSMOTE
Generates synthetic samples near decision boundary

ENN
Removes misclassified majority samples

Before:	90.9% vs 9.1%
After:	52.7% vs 47.3%

ISSUES
Complex pipeline, computational overhead, risk of over-cleaning valuable majority samples

IMPROVED APPROACH

TECHNIQUE

SMOTE

Synthetic Minority Over-sampling

Controlled Sampling Strategy
sampling_strategy = 0.3 (30% of majority)

k_neighbors = 3
Fewer neighbors for synthetic samples

Before:	90.9% vs 9.1%
After:	70% vs 30%

WHY BETTER
- Faster computation time
- Balanced but not oversampled
- Preserves original data patterns

MODELS COMPARISON

OLD MODELS

RandomForest

200 estimators, random_state=42

- Good feature importance extraction
- Robust to overfitting
- Limited interpretability

LogisticRegression

max_iter=1000

- Interpretable coefficients
- Fast training
- Linear decision boundaries

LIMITATION

No ensemble strategy, single model evaluation

IMPROVED MODELS

XGBoost

ADVANCED

n_estimators=300, max_depth=5, scale_pos_weight=5

- State-of-art gradient boosting
- Handles imbalanced data (scale_pos_weight)
- CUDA acceleration

IsolationForest

NOVEL

Anomaly detection, contamination=0.085

- Identifies high-risk patterns
- Unsupervised outlier detection
- Complementary to XGBoost

STACKING ENSEMBLE

Meta-learner: LogisticRegression combines both model predictions

ENSEMBLE STRATEGY EVOLUTION

OLD APPROACH

STRATEGY

NO ENSEMBLE

RandomForest | LogisticRegression

LIMITATIONS

- Single model predictions
- No diversity in predictions
- Higher variance risk
- No complementary strengths

Performance limited to single model capability

IMPROVED APPROACH

STRATEGY

STACKING ENSEMBLE

Base Model 1

IsolationForest (Anomaly Detection)

↓

Base Model 2

XGBoost (Classification)

↓

Meta-Learner

LogisticRegression (combines predictions)

WHY STACKING IS POWERFUL

- Learns optimal weight for each base model

PERFORMANCE METRICS

OLD PROJECT

ROC-AUC

0.72

Moderate discriminative ability

RECALL

0.68

Misses 32% of positive cases

PRECISION

0.71

29% false positive rate

Limited feature engineering and basic models constrain performance

IMPROVED PROJECT

ROC-AUC

0.86

+19% improvement

Excellent discrimination

RECALL

0.82

+21% improvement

Catches 82% of high-risk patients

PRECISION

0.79

+11% improvement

Fewer unnecessary interventions

KEY DRIVERS

Stacking ensemble + Risk_Score feature + XGBoost = Superior predictive power

SYSTEM INTELLIGENCE UPGRADE

CURRENT SYSTEM

CLASSIFICATION TYPE

BINARY

NO (0) VS YES (1)

BOTH VERSIONS

Same binary classification: No heart disease vs Heart disease

DECISION SUPPORT

Simple yes/no prediction limits clinical utility

LIMITATION

Cannot differentiate risk levels or prioritize patients

PROPOSED UPGRADE

CLASSIFICATION TYPE

3-CLASS SYSTEM

LOW RISK

Probability < 0.3

MODERATE RISK

0.3 ≤ Probability < 0.7

HIGH RISK

Probability ≥ 0.7

REAL-WORLD USEFULNESS

- Prioritize high-risk patients for immediate intervention
- Schedule moderate-risk patients for preventive care
- Reduce unnecessary tests for low-risk individuals
- Optimize healthcare resource allocation

STRENGTHS VS WEAKNESSES

OLD SYSTEM

- Comprehensive data exploration
- IterativeImputer for missing values
- RobustScaler for outliers
- Feature importance analysis

NEW SYSTEM

FOCUS

- **Risk_Score composite feature**
- **Stacking ensemble strategy**
- **XGBoost with hyperparameter tuning**
- **IsolationForest for anomaly detection**
- **+19% ROC-AUC improvement**
- **+21% Recall improvement**

TRADE-OFFS

PRECISION vs RECALL

- **High Precision:** Fewer false positives, reduced unnecessary treatments
- **High Recall:** Catches more at-risk patients, prevents missed cases

COMPLEXITY vs PERFORMANCE

New system is more complex but delivers significantly better performance. Worth the trade-off for clinical applications.

COMPUTATIONAL COST

Improved system requires more training time but maintains competitive inference speed suitable for production use.

FINAL CONCLUSION

WHY IMPROVED IS BETTER

- **19% higher ROC-AUC** → Better discrimination of heart disease risk
- **21% higher Recall** → Catches more at-risk patients
- **Risk_Score feature** → Leverages domain expertise
- **Stacking ensemble** → Combines multiple model strengths
- **Efficient pipeline** → 60% faster preprocessing

Thank You

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